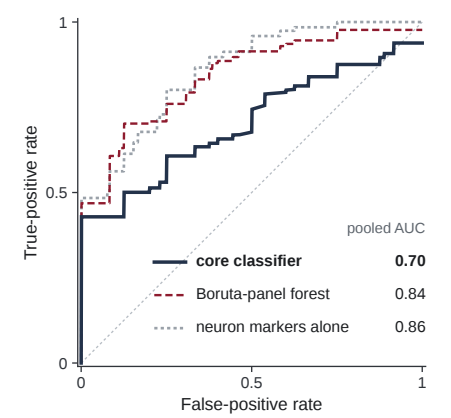
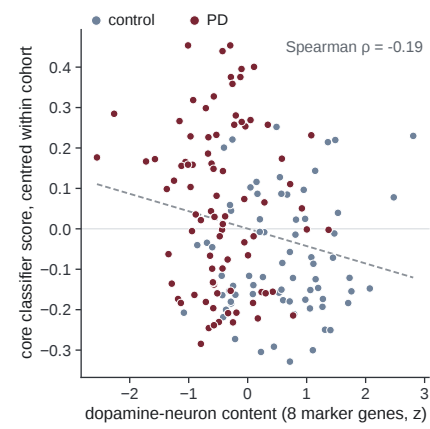


a Frozen models in eight independent bulk cohorts



8 cohorts, 158 people (72 control, 86 PD); trained only on the 63 laser-capture people
core classifier AUC 0.70 (95% CI 0.62–0.78), label shuffles within cohorts $P < 0.001$
curves: each cohort's ROC, averaged with weights equal to its PD–control pairs

b How much of it is neuron loss



	pooled AUC
neuron markers alone	0.86
core classifier	0.70
core, neuron content removed	0.62
panel forest	0.84
panel forest, neuron content removed	0.75
neurons alone, leave-one-out	0.75
neurons + core score, leave-one-out	0.76
neurons + panel score, leave-one-out	0.77

neuron content removed: score regressed on the neuron score within each cohort

c Every Boruta gene: effect in the laser-capture cohort and in each bulk cohort

